

POINT PETRE, ON, Canada

WMO: 714300

Lat: 43.8396N

Lon: 77.1530W

Elev: 79

StdP: 100.38

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -19.4	(c) -16.2	(d) -25.2	(e) 0.4	(f) -18.8	(g) -21.9	(h) 0.5	(i) -15.2	(j) 20.0	(k) -3.9	(l) 18.5	(m) -2.1	(n) 5.6	(o) 30	(p) 0.678

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 6.7	(c) 26.4	(d) 22.7	(e) 25.3	(f) 21.9	(g) 24.4	(h) 21.3	(i) 23.6	(j) 25.5	(k) 22.7	(l) 24.6	(m) 21.9	(n) 23.8	(o) 4.5	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 22.9	(b) 17.8	(c) 24.9	(d) 22.0	(e) 16.8	(f) 24.1	(g) 21.1	(h) 15.9	(i) 23.4	(j) 70.8	(k) 25.7	(l) 67.3	(m) 24.8	(n) 64.4	(o) 23.9	(p) 27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 16.2	(b) 14.2	(c) 12.6	(e) -23.0	(f) 28.2	(g) 3.0	(h) 1.7	(i) -25.2	(j) 29.4	(k) -26.9	(l) 30.4	(m) -28.6	(n) 31.4	(o) -30.9	(p) 32.6		
(a) 16.2	(b) 14.2	(c) 12.6	(e) -23.3	(f) 24.4	(g) 3.0	(h) 1.3	(i) -25.5	(j) 25.3	(k) -27.3	(l) 26.1	(m) -28.9	(n) 26.8	(o) -31.1	(p) 27.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.0	-4.8	-3.8	-0.2	5.0	10.7	15.9	20.2	20.5	16.9	10.7	4.8	-0.7	(6)
(7)		DBStd	9.69	6.34	5.03	4.49	3.18	3.16	2.98	2.50	2.55	3.52	3.98	4.25	5.17	(7)
(8)		HDD10.0	1879	459	388	317	152	29	1	0	0	1	40	162	331	(8)
(9)		HDD18.3	3954	718	621	576	399	237	83	11	9	66	237	407	589	(9)
(10)		CDD10.0	1148	0	0	1	2	51	178	316	326	209	63	4	0	(10)
(11)		CDD18.3	182	0	0	0	0	1	11	68	77	24	1	0	0	(11)
(12)		CDH23.3	481	0	0	0	0	1	14	187	229	49	0	0	0	(12)
(13)		CDH26.7	22	0	0	0	0	0	1	8	11	2	0	0	0	(13)
(14)	Wind	WSAvg	6.0	7.6	7.1	6.3	6.0	5.0	4.4	4.4	4.8	5.4	6.7	7.2	7.7	(14)
(15)	Precipitation	PrecAvg	893	66	55	63	79	79	79	86	79	80	85	72	71	(15)
(16)		PrecMax	1089	143	104	114	148	145	149	192	126	137	160	102	124	(16)
(17)		PrecMin	705	36	20	23	24	27	19	33	31	38	27	25	38	(17)
(18)		PrecStd	95	25	19	24	31	33	37	43	24	28	35	22	20	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.2	5.8	11.6	15.1	21.6	25.1	27.7	28.0	26.2	20.6	14.3	11.0	(19)
(20)			MCWB	6.4	4.6	8.6	10.4	17.0	20.7	23.2	24.4	22.8	18.6	12.8	10.0	(20)
(21)		2%	DB	5.7	4.5	8.1	12.3	19.0	23.2	26.3	26.5	24.7	19.2	13.0	9.1	(21)
(22)			MCWB	4.7	3.4	5.9	8.6	15.6	19.8	22.5	23.0	21.7	17.1	11.3	8.1	(22)
(23)		5%	DB	4.5	3.5	6.4	10.6	17.0	21.7	25.2	25.5	23.4	17.9	11.9	7.5	(23)
(24)			MCWB	3.6	2.4	4.7	7.6	13.9	18.9	21.9	22.1	20.7	15.9	10.5	6.3	(24)
(25)		10%	DB	3.2	2.6	5.1	9.2	15.2	20.5	24.3	24.6	22.3	16.7	10.8	6.0	(25)
(26)			MCWB	2.3	1.5	3.6	6.8	12.7	17.9	21.1	21.4	19.8	14.8	9.2	4.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.7	4.8	9.8	11.8	18.4	22.1	24.5	25.2	23.8	19.3	13.4	10.2	(27)
(28)			MCDWB	7.2	5.5	11.4	13.5	20.8	24.0	26.3	26.9	25.5	20.1	14.1	10.6	(28)
(29)		2%	WB	5.0	3.6	6.4	9.5	16.2	20.6	23.4	23.8	22.4	17.4	11.8	8.3	(29)
(30)			MCDWB	5.6	4.3	7.5	11.2	18.3	22.4	25.4	25.8	24.1	18.8	12.6	9.0	(30)
(31)		5%	WB	3.7	2.6	5.0	8.2	14.5	19.4	22.5	22.9	21.3	16.3	10.6	6.4	(31)
(32)			MCDWB	4.4	3.3	6.1	10.1	16.5	21.2	24.6	24.8	22.9	17.6	11.7	7.4	(32)
(33)		10%	WB	2.3	1.6	3.7	7.0	13.1	18.2	21.7	22.1	20.2	15.0	9.4	4.9	(33)
(34)			MCDWB	3.1	2.4	4.9	8.7	14.8	20.0	23.8	24.0	22.0	16.5	10.6	5.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.2	6.7	6.1	6.1	6.4	6.4	6.7	7.0	5.9	5.8	6.2	(35)	
(36)			MCDBR	5.9	5.2	6.7	8.3	8.8	7.7	7.2	7.0	7.1	5.7	5.9	5.7	(36)
(37)			MCWBR	6.0	4.9	5.3	5.7	5.9	5.0	4.8	4.8	5.0	4.9	5.4	5.6	(37)
(38)		5% WB	MCDBR	5.7	5.1	6.4	7.4	7.8	6.7	6.4	6.2	6.3	5.5	5.8	5.4	(38)
(39)			MCWBR	5.9	5.0	5.4	5.6	5.9	4.7	4.7	4.6	4.8	5.0	5.6	5.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.298	0.305	0.321	0.363	0.387	0.427	0.427	0.416	0.375	0.352	0.311	0.288	(40)	
(41)		taud	2.431	2.401	2.434	2.382	2.371	2.290	2.300	2.317	2.413	2.460	2.523	2.511	(41)	
(42)		Ebn at Noon	832	890	920	902	885	847	842	838	849	825	815	816	(42)	
(43)		Edh at Noon	74	90	100	114	119	130	127	121	102	85	68	63	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.28	2.08	3.42	4.51	5.51	5.86	5.96	5.25	4.18	2.47	1.52	1.00	(44)	
(45)		RadStd	0.13	0.19	0.30	0.49	0.54	0.47	0.47	0.28	0.34	0.22	0.19	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (37 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+74	N/A

Nomenclature: See separate page